

Problem Statement

Bridging the Tokenization Gap in African LLMs via SuperBPE — TRI AI Saturdays, Cohort 10

Millions of speakers of African languages — including users, developers, and researchers working in Swahili, Hausa, Yoruba, Amharic, and related languages — are structurally disadvantaged by AI systems built around English. Because mainstream tokenizers are trained on English-dominated text, they do not recognize the morphological structure of morphologically rich or agglutinative African languages, where a single word can carry the meaning of an entire English sentence through layered prefixes, infixes, and suffixes. Words in these languages are broken into small, often meaningless fragments instead of the coherent units a native speaker would recognize, inflating the computational cost of using AI in these languages and degrading result quality for their speakers. Left unaddressed, this deepens the digital divide and keeps high-quality AI access concentrated among wealthier, English-speaking institutions.

This project addresses the problem by building a tokenization pipeline trained directly on native corpora across nine African languages spanning three major language families — Niger-Congo Bantu languages (Swahili, Chichewa, Zulu, Kinyarwanda), Niger-Congo Volta-Niger and West Atlantic languages (Yoruba, Wolof, and Igbo, added as a documented bonus ninth language), and Afro-Asiatic languages (Hausa, and Amharic in its Ethiopic script) — rather than adapting an English-centric tokenizer after the fact. We evaluate SuperBPE, a two-stage byte-pair-encoding scheme that learns ordinary subwords before learning frequent multi-word “superwords,” against a matched standard byte-pair-encoding baseline, to test whether it further reduces the fragmentation this project exists to address.

Guiding Values

Inclusivity. The project spans three distinct language families rather than concentrating on the handful of most widely spoken African languages, reflecting a commitment to representing linguistic diversity rather than the languages easiest to source data for.

Cultural Preservation. Technical choices were made to treat each language's structure as worth preserving rather than normalizing away: tonal and diacritic marking in Yoruba and Igbo is retained as meaningful signal rather than stripped as noise, and Amharic's Ethiopic script is handled as a first-class writing system throughout the pipeline rather than an edge case to work around.

Digital Equity. The project's central technical aim — reducing how many tokens are needed to represent African-language text — is a direct, measurable step toward closing the compute-cost and inference-speed gap between English and these languages, since token count drives both.